

SUMMARY

Innovative Lead Software Engineer with extensive experience in managing software development projects. Leverages strong leadership and effective communication to drive innovation and efficiency while adhering to tight schedules and budgets. Committed to continuous learning and community betterment, with a focus on applying science and technology to create practical solutions that enhance productivity.

SKILLS

- Backend design
- Microservices architecture
- API design
- Node.js
- JavaScript
- PostgreSQL
- vector Db
- NEO4J
- Generative AI & AI Engineering
- NoSQL
- Redis
- Database tuning
- Kubernetes
- Linux
- Google Cloud
- Git

EXPERIENCE

LEAD SOFTWARE ENGINEER | 06/2023 to Current

falabella - Bangalore, India

- Led backend architecture and scaling initiatives for high-traffic APIs handling up to 15K TPS by redesigning PostgreSQL schemas, flattening JSONB-heavy data structures, and optimizing query/index performance.
- Designed and reviewed backend architectures for microservice-based systems using Node.js, PostgreSQL, Redis, Neo4j, and GCP services.
- Split a shared backend service into two independent microservices to enable separate scaling, deployments, and ownership based on different business workflows.
- Improved system reliability and reduced latency by identifying bottlenecks through query analysis, monitoring, logging, and production troubleshooting.
- Reduced query latency and infrastructure load through database tuning, optimized data models, caching strategies, and query refactoring.
- Owned critical production incidents, conducted RCA, and implemented long-term fixes, significantly improving system stability and operational reliability.
- Mentored team members in system design, production debugging, code reviews, and development of scalable backend services, enhancing overall team capability.
- Collaborated with product and business stakeholders to translate functional requirements into scalable, maintainable backend solutions, aligning technical outcomes with business needs.

SENIOR SOFTWARE ENGINEER | 07/2021 to 06/2023

falabella - Bangalore, India

- Led development of a payment gateway microservice platform that exposed standardized APIs for multiple businesses to integrate with different payment providers.
- Integrated and managed end-to-end payment flows with PSPs including PayU, Fpay, Servipag, and Pago Efectivo, supporting multiple payment methods across regions.
- Worked extensively with GCP services including Google Pub/Sub, Cloud Storage, Cloud Tasks, and Cloud Functions to build reliable and event-driven payment workflows.
- Built and delivered payment features such as wallet integrations and partial refunds, improving payment experience and operational workflows.
- Collaborated with product, QA, and platform teams in an Agile environment to deliver production-ready features and enhance system performance.
- Contributed to scalable microservice-based backend systems using Node.js/JavaScript, PostgreSQL, Docker, and Kubernetes.
- Participated in system design discussions and production debugging, contributing to reliability improvements for payment and transaction processing systems.

ASSOCIATE SOFTWARE DEVELOPER | 06/2017 to 06/2021

SCIENTIFIC GAMES - Bangalore, India

- Developed full stack applications, enhancing user experience and functionality.
- Experienced in Node.js, Express.js, vanilla Javascript, redis.
- Troubleshoot and resolved complex software issues, ensuring system stability and reliability.
- Utilized Git for version control, streamlining collaboration across development team.

EDUCATION

Vellore Institute of Technology - Vellore | 05/2017

BTech - Information technology

HEHS - TINSUKIA

High School Diploma

HOBBIES AND INTERESTS

- Music
- Travelling
- Experimenting
- Physics
- Spirituality

WEBSITES & SOCIAL LINKS

- GitHub: <https://github.com/kalpagarwal>
- LinkedIn: <https://www.linkedin.com/in/kalp-agarwal-433123109/>

PERSONAL PROJECTS

AI Learning

- Built a personal AI learning project using Node.js, TypeScript, Next.js, PostgreSQL, and Qdrant to understand RAG, AI orchestration, memory, and tool-calling workflows.
- Implemented document ingestion, chunking, embeddings, vector search, and conversational Q&A using a RAG-based retrieval pipeline.
- Added multiple agents for tutoring, quiz generation, planning, and research workflows with a simple orchestration layer.
- Integrated tool-calling flows, conversation memory, quiz tracking, and basic observability/debug tracing to better understand AI execution flows.
- Structured the backend into separate modules for retrieval, orchestration, memory, tools, and agent execution for easier experimentation and learning.
- Experimented with local LLMs and embedding models using LM Studio, including Qwen2.5-3B-Instruct and nomic-embed-text-v1.5, through OpenAI-compatible APIs.